

Solution Requirements

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Name	FIT BUDDY AI
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users provide a User ID and profile details. The documented project does not implement <i>[Describe how users will password login or identity verification. securely log in or verify their identity]</i>	Low
2	Authorization levels	User-facing access is described, along with an <i>[Define the user roles and admin view for all users and plans. A separate authentication or role-enforcement what access/permissions mechanism is not documented. each role has]</i>	Medium

3	External interfaces	Integrates Google Gemini 1.5 Pro for workout <i>[Detail any third-party APIs, generation and plan updates, Gemini Flash for nutrition/recovery tips, and SQLite with external databases, or SQLAlchemy for persistence.</i>	High
		The Gemini and SQLite integrations described <i>services the system must</i> above continue in this row. Data is exchanged through the FastAPI backend. <i>connect to]</i>	High
4	Transactions processing	Processes submissions to generate a 7-day workout plan and nutrition tip, accepts <i>[Explain how data feedback, revises the original plan with exchanges, payments, or Gemini, and stores the revised plan. No payment processing is documented. core system transactions will be handled]</i>	High
5	Reporting	Provides an admin dashboard listing registered users and their original/updated <i>[Describe the types of workout plans. No report export function is reports, dashboards, or data documented. exports the system must generate]</i>	Medium
6	Business rules, etc.	Generates a structured 7-day plan using age, weight, goal and intensity; includes warm-up. <i>[List specific logic, main workout and cooldown/recovery constraints, or workflows guidance. Feedback revises the existing plan while preserving the original. unique to the business process]</i>	High
7	Compliance to laws or regulations	The project document does not specify a <i>[Specify any legal, industry, particular legal, industry, or privacy regulation such as GDPR or HIPAA. Compliance or data privacy standards requirements are therefore not defined in the supplied documentation. the system must adhere to (e.g., GDPR, HIPAA)]</i>	Low

8	Other	Provides a concise nutrition or recovery tip [Add any additional aligned with the selected fitness goal and displays it with the generated workout plan. <i>functional requirements here]</i>	Medium
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Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed Performance & Speed	[The application is designed for real-time web interaction and invokes Gemini APIs during plan generation and feedback updates. <i>Describe response times, processing speed, etc.]</i>	(No numeric response-time target is documented. <i>e.g., Page loads in < 2 seconds</i>)
2	Scalability	The architecture is modular, using FastAPI, Gemini APIs and SQLite. The supplied project document describes future cloud deployment but gives no capacity target. <i>Describe how the system handles growth in users or data]</i>	No maximum concurrent-user or (data-volume target is specified. <i>e.g., Supports up to 10,000 concurrent users</i>)
3	Security & Data Privacy Security & Data Privacy	[The Gemini API key is stored through environment variables. User and plan data are stored in SQLite. Specific encryption and vulnerability-testing requirements are not documented. <i>Detail data protection, encryption, and vulnerability management]</i>	(API key should not be hard-coded; additional security controls are not specified. <i>e.g., End-to-end AES-256 encryption)</i>
4	Reliability & Availability Reliability & Availability	[The application is deployed locally with Uvicorn and provides FastAPI interactive API documentation. No uptime percentage or fault-tolerance target is documented. <i>Specify uptime requirements and fault tolerance]</i>	(Successful local startup and access to documented pages/endpoints. <i>e.g., 99.9% uptime per month</i>)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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5	Usability & Accessibility Usability & Accessibility	[The UI uses HTML, CSS and Jinja2, with Flexbox layouts, mobile media queries, labeled Describe user experience standards inputs and consistent responsive result pages. and accessibility compliance]	(Users can submit details, view plans and submit feedback e.g., Complies with WCAG through the documented interface. 2.1 AA standards)
6	Other Other	The code is organized into separate modules for [routes, AI generation, database access and Add any additional non-functional Jinja2 templates, supporting maintainability and requirements local deployment. (e.g., Maintainability, Portability)]	Documented modules and templates remain separable and the application runs in the stated Python/FastAPI environment.